

Aman Palanati

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EDUCATION

University of California, San Diego

La Jolla, CA

Bachelor of Science in Computer Engineering | GPA: 3.915

Sep. 2024 – June 2027

Coursework: Data Structures and Object-Oriented Design, Discrete Mathematics, Systems Programming and Software Tools, Computer Organization, Analog Circuit Design, Circuits and Systems, Components and Circuits Laboratory

EXPERIENCE

Undergraduate Research Assistant

Oct. 2024 – June 2025

UC San Diego Engineers for Exploration

La Jolla, CA

- Contributed to the development of **Smartfin**, a surfboard-mounted, sensor-integrated device that collects and remotely stores coastal ocean data, through iterative **Agile** development cycles
- Developed embedded firmware in **C++** to interface with numerous environmental sensors
- Implemented **GitHub** for version control, structured documentation, and collaboration with team members
- Designed structural components using CAD software, including **Onshape** and **Fusion 360**

Robotics Tutor

Aug. 2020 – Feb. 2024

FIRST Robotics

Chesterfield, MO

- Conducted weekly sessions to guide students in building and programming robots using **LEGO Mindstorms**
- Taught fundamental programming concepts like **sensor integration** and **autonomous navigation** using **Python**
- Provided mentorship on mechanical design principles like **gear ratios** and **structural stability**

PROJECTS

Sonata | *React.js, TypeScript, Flask, Python, Supabase, CSS, GitHub*

July 2025 – Present

- Architected and developed a **full-stack web application** to connect music teachers with students using **React.js**, **TypeScript**, **Flask**, and **Python**, enabling dynamic navigation, protected routes, and role-based access control
- Built multi-step forms with **React Hook Form** and **Yup validation**, incorporating **real-time error handling**, and reusable **custom hooks** for field management
- Implemented secure **authentication** with **Supabase Auth** and **Google OAuth**, including session management, password reset workflows, and custom route guards
- Designed a modular **RESTful API** backend with **dependency injection**, integrating **cloud-hosted PostgreSQL** services for user authentication and file storage, with support for **CORS** and secure session lifecycle management

EXTRACURRICULARS

IEEE - Eta Kappa Nu (HKN) | *Member*

Jan. 2026 – Present

- Selected for the international honor society for **IEEE**, participating in **professional networking**, **community outreach**, and **technical development** within the UC San Diego engineering community

Association for Computing Machinery | *Member*

Sep. 2024 – Present

- Participated in **workshops** covering topics such as **artificial intelligence** and **cybersecurity**
- Attended **hackathons**, developing software solutions and refining coding skills in a **team setting**

FIRST Tech Challenge | *Team Captain*

Sep. 2020 – May 2024

- Developed **computer vision** algorithms in **Java** using **TensorFlow Object Detection**, allowing the robot to identify and interact with elements autonomously
- Implemented **PID control systems** to enhance autonomous navigation and task execution
- Conducted debugging and optimization with **sensor calibration**, **real-time logging**, and **iterative testing**
- Designed optimal **3D-printed** and **laser-engraved** robot components using **Onshape**

SKILLS

Languages & Frameworks: Python, Java, C, C++, JavaScript/TypeScript, HTML/CSS, Flask, React.js, PostgreSQL, GCP

Development Skills: JUnit/PyTest, Code Review, Git/GitHub, Unix/Linux, Data Structures, Cloud Services, Database Design

Engineering & CAD: Circuit Design/Analysis, Circuit Components, Onshape, Fusion 360

Soft Skills: Leadership, Problem-Solving, Teamwork